Project Requirement Document: Architecture & Components

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1. Component & Credentials List

System Components

Category	Component / Integration	Purpose
Frontend	React	Accepts user text and renders the AI's chat responses.
Orchestration	n8n Node Group (Webhook, AI Agent, Supabase, Respond)	Manages API routing, database queries, and AI chaining.
Data Store	Supabase (pgvector)	PostgreSQL database storing embedded company documents.
AI Models	LLM API & Embedding Model	Synthesizes answers and converts text into vector coordinates.

Required Credentials

• React Frontend: Header Auth account / API Key (for securing the intake webhook)

• Database: Supabase API URL & Service Role Key / Postgres Database URL

• AI Pipeline: LLM API Key (e.g., Groq, OpenAI, or Gemini) & Embedding Model API Key

• Email Routing: Gmail OAuth / SMTP credentials (for HR escalation)

2. Security Considerations

• Credential Management: All API keys (LLM, database) are securely stored in n8n's encrypted credential manager, ensuring they are never exposed in the payload.

• CORS Protection: The n8n Webhook restricts incoming requests exclusively to the React application's specific domain or localhost port.

• Data Isolation: The database connection utilized for the chatbot workflow has strictly read-only permissions to prevent accidental overwrites or SQL injection attacks.

3. Failure Handling

• Hallucination Prevention: If the vector search returns zero relevant results, the system defaults to a strict "Unknown-answer" prompt to prevent the AI from fabricating policies.

• Escalation Routing: Unanswered queries automatically trigger a Switch Node branch to a secondary workflow that drafts a support ticket or sends an email to HR.

• Timeout Triggers: Global n8n error-trigger nodes are configured to alert the system administrator if the LLM API experiences an outage or rate limit.